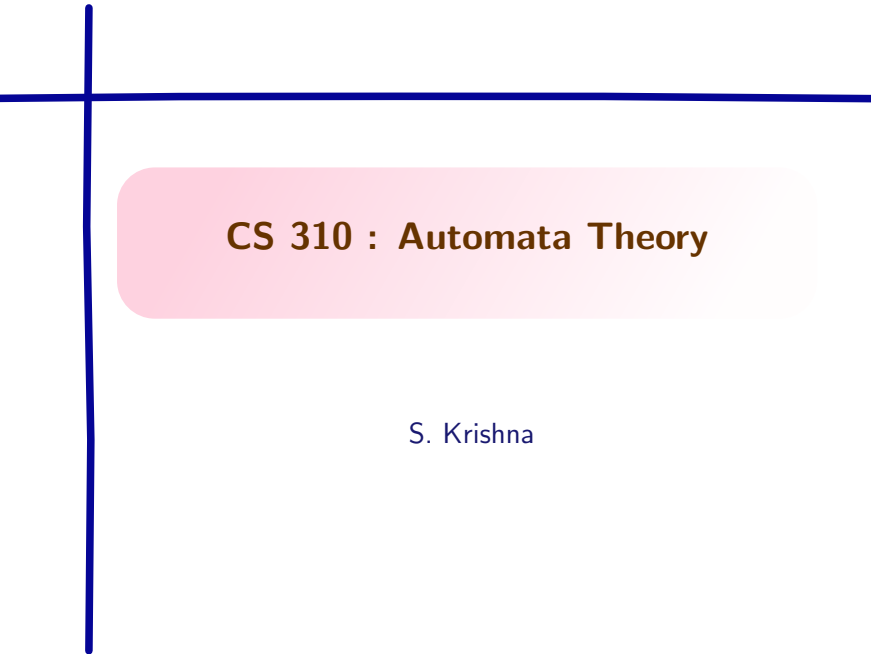




CS 310 : Automata Theory

S. Krishna

How to use pumping lemma?

Recall : Pumping Lemma as an Implication

L is regular $\Rightarrow$

$$\exists p \forall w \in L. [|w| \geq p \Rightarrow \exists x, y, z (w = xyz \wedge y \neq \epsilon \wedge |xy| \leq p \wedge (\forall i \geq 0, xy^i z \in L)))]$$

Recall : Its contrapositive

$$\begin{aligned} \forall p \exists w \in L. [|w| \geq p \wedge \forall x, y, z ((w = xyz \wedge y \neq \epsilon \wedge |xy| \leq p) \\ \Rightarrow \exists i \geq 0. xy^i z \notin L)] \\ \Rightarrow L \text{ is not regular} \end{aligned}$$

Contrapositive of pumping lemma

In words,

Theorem

If

- ▶ for all p ,
- ▶ there is a word $w \in L$ such that $|w| \geq p$,
- ▶ for each breakup of w into three words $w = xyz$ such that
 1. $y \neq \epsilon$ and
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We have proven that L is not regular.

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- ▶ Clearly, $0^{p-j} 1^p \notin L$. Therefore, L is not regular.

Example 3: using pumping lemma

Let word w^R be reverse of word w .

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- ▶ Let $k = 2$ for each i and j .

$$0^i (0^j)^2 0^{p-j-i} 1 10^p = 0^{p+j} 1 10^p \notin L_{rev}$$

Properties of regular languages

Exercises:

1. Is subset of any regular language regular?
2. Is superset of any regular language regular?
3. If $L_1 \cup L_2$ is regular, then are L_1 and L_2 regular?
4. Is this regular: $\{0^n 1^m \mid n \neq m\}$?
5. Is this regular: $\{0^n 1^m \mid n > m\}$?
6. Is this regular: $\{0^n 1^m \mid n \text{ is odd and } m \text{ is even}\}$
7. Is this regular: $\{0^n 1^{n+1} \mid n \geq 0\}$

Example 4 : not using pumping lemma

Using regularity properties

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- ▶ Suppose L is regular. Note that $\Sigma = \{0, 1\}$.
- ▶ Consider the language $L' = \bar{L} \cap 0^* 1^*$, where $\bar{L}$ is the complement of L .
- ▶ This must be regular by closure properties.
- ▶ But $L' = \{0^n 1^n \mid n \in \mathbb{N}\}$, which we know is not regular.
- ▶ Hence we have a contradiction and L is not regular.

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- ▶ This completes the proof.

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- ▶ So, $|xy^kz| = (n - j) + kj$.
- ▶ Let $k = n - j$. Therefore, $|xy^kz| = (n - j)(1 + j)$.
- ▶ Since both $(n - j) > 1_{(\text{why?})}$ and $1 + j > 1_{(\text{why?})}$, $xy^kz \notin L_{\text{prime}}$.

More examples using pumping lemma

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- ▶ Let $k = 2$. Therefore, $|xy^kz| = p^2 + j$.
- ▶ Since $0 < j \leq p$, $p^2 < p^2 + j < p^2 + 2p + 1$.

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- ▶ Since $0 < j \leq p$, $p^2 < p^2 + j < p^2 + 2p + 1$.
Therefore, $xy^2z \notin L$.

Pumping Lemma and Non-Regularity

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Pumping Lemma and Non-Regularity

Pumping Lemma is a necessary but not sufficient condition for regularity.

$$F = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } (i = 1 \Rightarrow j = k)\}$$

- ▶ F is **not** regular
- ▶ F **satisfies** the pumping lemma

$F = \{a^i b^j c^k \mid i, j, k \geq 0, (i=1 \Rightarrow j=k)\}$ not regular

Suppose F were regular.

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$R = ab^*c^*$ is regular, intersect with R

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If F is regular, $F \cap R$ must be regular too

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So it suffices to show $L = \{ab^n c^n \mid n \geq 0\}$ is not regular

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Suppose it were regular. Let p be its pumping length.

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- ▶ Hence, x, y consist only of a 's and/or b 's.

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- ▶ Take $i = 0$: $xy^0z = xz$ removes the $|y| > 0$ symbols of y from the a/b -prefix, while leaving all p c 's untouched.

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- ▶ So xz has fewer than p a 's and b 's combined (or lost its single a) but has exactly p c 's $\Rightarrow xz \notin L$.

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So L , and hence F , is not regular

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“ $\exists p$ such that for all $w \in L$, $|w| \geq p, \dots$ ”

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Take pumping length $p = 2$.

$$F = \{a^i b^j c^k \mid i, j, k \geq 0, (i=1 \Rightarrow j=k)\}$$

F 's only constraint is at $i = 1$. For $i \neq 1$, j, k can be anything.

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Strategy

Pump only inside the a 's, chosen so the count of a is not 1

Case analysis: $i = 0$ and $i = 1$

Let $w = a^i b^j c^k \in F$, $|w| \geq 2$.

- ▶ $i = 0$: take $x = \varepsilon$, $y =$ first symbol of w , $z =$ rest.

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Pumping only changes the b or c count; $i \neq 1 \Rightarrow$ is in F .

- ▶ $i = 1$: then $j = k$. Take $x = \varepsilon$, $y = a$, $z = b^j c^k$. $xy^\ell z = a^\ell b^j c^k$.
 - ▶ At $\ell = 1$: in F as $j = k$
 - ▶ For $\ell \neq 1$: a count $\neq 1 \Rightarrow$ is in F .

Case analysis: $i = 2$ and $i \geq 3$

Let $w = a^i b^j c^k \in F$, $|w| \geq 2$.

- ▶ $i = 2$: take $x = \varepsilon$, $y = aa$, $z = b^j c^k$.

Case analysis: $i = 2$ and $i \geq 3$

Let $w = a^i b^j c^k \in F$, $|w| \geq 2$.

► $i = 2$: take $x = \varepsilon$, $y = aa$, $z = b^j c^k$.

$xy^\ell z = a^{2\ell} b^j c^k$. The a -count 2ℓ is always **even** $\Rightarrow \neq 1 \Rightarrow$ is in F

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► $i \geq 3$: take $x = a$, $y = a$, $z = a^{i-2} b^j c^k$ (note $|xy| = 2 \leq p$).

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► $i \geq 3$: take $x = a$, $y = a$, $z = a^{i-2} b^j c^k$ (note $|xy| = 2 \leq p$).

$xy^\ell z = a^{(i-1)+\ell} b^j c^k$.

Since $i \geq 3$: a -count $= i-1 + \ell \geq 2$ for all $\ell \geq 0 \Rightarrow \neq 1 \Rightarrow$ is in F .

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Since $i \geq 3$: a -count $= i-1 + \ell \geq 2$ for all $\ell \geq 0 \Rightarrow \neq 1 \Rightarrow$ is in F .

Every case yields a valid pumping decomposition. So F satisfies the pumping lemma, despite being non-regular.

Takeaway

What this example shows

The pumping lemma can only **disprove** regularity, it can never **prove** it, and it cannot **always** detect non-regularity either.

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To show F is non-regular, we used **closure under intersection** with a simpler regular language, reducing to the known non-regular $\{ab^nc^n \mid n \geq 0\}$.

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What this example shows

The pumping lemma can only **disprove** regularity, it can never **prove** it, and it cannot **always** detect non-regularity either.

To show F is non-regular, we used **closure under intersection** with a simpler regular language, reducing to the known non-regular $\{ab^nc^n \mid n \geq 0\}$.

If the pumping lemma seems to “work” for a language, that’s not proof of regularity.

Where does the pumping lemma fall short?

We consider a split $w = xyz$ with $|xy| \leq p$ the pump window is anchored at a prefix, trying to find a bad word.

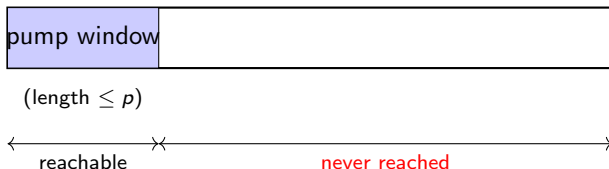
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- ▶ For F , *exactly one* a forces $j = k$ is at the front, used by the pump window
- ▶ the pumping lemma only needs **one** safe split to exist
- ▶ we can pick the **pump a** option, which is the safe $i \neq 1$ case,
- ▶ the unsafe option (pumping the b 's) is never forced